

Phase 8H — Dynamic Rainfall & Risk Fusion Layer

Technical Walkthrough, CWC Telemetry Integration & Deterministic Fusion Matrix

Telemetry Source: Central Water Commission (CWC, 73 stations)	Spatial Cap: 50.0 km radius threshold
Freshness SLA: 6.0 hours (age > 6h flagged STALE)	Macro Context: IMD State/District Administrative Bulletins
Authoritative Decision: Discrete Risk Tier (LOW/WATCH/HIGH/CRITICAL)	Test Suite: 40 / 40 Tests Passed

1. Deterministic Decision Matrix

Static Susceptibility Tier	NORMAL Rain	WATCH Rain	HIGH Rain	NO_DATA Rain
LOW	LOW	WATCH	WATCH (Damped)	LOW
MODERATE	LOW	WATCH	HIGH	WATCH (Precautionary)
HIGH	WATCH (Baseline)	HIGH	CRITICAL	WATCH (Unobserved)
VERY_HIGH	WATCH (Baseline)	HIGH	CRITICAL	HIGH (Unobserved)

Operational Score Semantics: Metric Name: <code>operational_fusion_score</code> (with backwards-compatible alias <code>risk_score</code>). Semantics: Engineering ordering synthesis for spatial ranking and map sorting; NOT a calibrated statistical event probability. Authoritative Output: The discrete categorical <code>risk_level</code> (LOW, WATCH, HIGH, CRITICAL).
--

2. Scientific Boundary Enforcements

- Unobserved Rainfall:** Returned as explicit null and never converted to 0.0 mm.
- Spatial Capping:** CWC stations > 50 km discarded (status: NO_RELIABLE_LOCAL_STATION).
- Freshness SLA:** Telemetry older than 6 hours flagged STALE.
- IMD Honesty:** Retained at administrative district/state scope; never fabricated as localized point telemetry.